

Independent Surreal Copies and Arithmetic Intersections

Prescribed Hahn cores, algebraic independence,
and two transcendental descent obstructions

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Abstract

Let H be any set-sized divisible additive subgroup of the surreal numbers, and let $L = \mathcal{K}_{\mathbb{R}}(H)$ be the full Conway–Hahn field with exponents in H . We construct an ordinal-indexed proper class of strong, real-fixing, monomial-preserving self-embeddings of $\mathbf{No}$ whose images are jointly linearly disjoint over L and have pairwise intersection exactly L . All embeddings fix L pointwise and preserve and reflect the omnific integers. The same construction gives complex-fixing, conjugation-compatible copies of $\mathbf{No}(i)$ and its Gaussian omnific ring. Every set of parameters can be fixed by choosing a suitable H .

The proof combines a cut-realization lemma that avoids an arbitrary set of previously chosen values with a coset-slice criterion for linear disjointness of full Hahn fields. It yields exact Boolean meet diagrams and descent of all polynomial relations across their squares. Two consequences distinguish this construction from mere universality. First, the fraction field of the intersection of two full omnific copies is a proper subfield of their common field, with a transcendental gap for every set-sized H . For $H = \mathbb{Q}$ a witness is $1 + \sum_{n \geq 1} \omega^{-n!}$; real constants already belong to the smaller fraction field. Second, explicit independent copies with intersection $\mathbb{R}$ have a full Hahn join containing an explicitly described series transcendental over their ordinary compositum. Both obstructions persist on complexification.

Status and scope. The article provides complete mathematical arguments relative to standard Conway normal form and Hahn-field foundations. The proposed new contribution is the combined prescribed-intersection theorem and its arithmetic and compositum applications. The ingredients and the bounded-support transcendence mechanism are not claimed as new. This is an unrefereed, non-formalized draft, not a certified priority claim or a claimed solution of a named longstanding conjecture.

Repository comparison is pinned to commit

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Sources, limitations, and a concrete research program are included.

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1 Three questions and their answers

The existence of an embedding into the surreal numbers says little about how two embedded copies meet. An intersection question asks for much more: which elements must be shared, which algebraic relations can pass from one copy to another, and whether their distinguished integer rings reconstruct the same common field. This article treats these questions for *full proper-class copies* of the surreal field, rather than only for small subfields.

The ambient presentation matters. A surreal has a unique Conway normal form

$$x = \sum_{g \in S} r_g \omega^g,$$

where $S \subseteq \mathbf{No}$ is a reverse-well-ordered *set*. For an additive subgroup $G \subseteq \mathbf{No}$, put

$$\mathcal{K}_{\mathbb{R}}(G) = \{x \in \mathbf{No} : \text{supp}(x) \subseteq G\}, \quad \mathcal{K}_{\mathbb{C}}(G) = \mathcal{K}_{\mathbb{R}}(G)(i).$$

These are full Hahn fields: there is no additional bound on the cardinality or order type of a set-sized support. If G is divisible, the real field is real closed and the complex field is algebraically closed. These foundations are classical; see [1, 2, 3].

1.1 Prescribing the common field

Theorem A (Independent full copies over a prescribed Hahn core). *Let $H \subseteq \mathbf{No}$ be a set-sized divisible additive subgroup and write $L_k = \mathcal{K}_k(H)$ for $k = \mathbb{R}, \mathbb{C}$. There are class maps*

$$\Phi_{\alpha} : \mathbf{No} \longrightarrow \mathbf{No} \quad (\alpha \in \mathbf{On})$$

with the following properties.

- (i) *Each Φ_{α} is a proper ordered field embedding, fixes $L_{\mathbb{R}}$ pointwise, sends Conway monomials to Conway monomials, and preserves and reflects Hahn summability of set-indexed families.*

- (ii) *With $K_{\alpha} = \Phi_{\alpha}(\mathbf{No})$,*

$$K_{\alpha} \cap K_{\beta} = L_{\mathbb{R}} \quad (\alpha \neq \beta).$$

Every finite subfamily is jointly linearly disjoint over $L_{\mathbb{R}}$.

- (iii) *For $R_{\alpha} = \Phi_{\alpha}(\mathbf{Oz})$,*

$$R_{\alpha} = K_{\alpha} \cap \mathbf{Oz}, \quad R_{\alpha} \cap R_{\beta} = L_{\mathbb{R}} \cap \mathbf{Oz}.$$

Moreover $\Phi_{\alpha}^{-1}(\mathbf{Oz}) = \mathbf{Oz}$.

- (iv) *The coefficientwise complexifications have the same assertions over $L_{\mathbb{C}}$, commute with conjugation, and preserve and reflect $\mathbf{Oz}[i]$.*

The proof is given in [sections 3 to 6](#). “Jointly linearly disjoint” is stronger than pairwise intersection being small: finitely many independent choices from different copies satisfy no new polynomial relations over the common core. Proper-class terminology is made precise in [section 2](#); no tensor product indexed by a proper class is required.

A useful immediate specialization is $H = \{0\}$. It produces full copies of $\mathbf{No}$ intersecting exactly in $\mathbb{R}$, while their omnific rings intersect exactly in $\mathbb{Z}$. A second specialization, $H = \mathbb{Q}$, fixes not just ω and its rational powers but every Hahn series with rational exponents. In either case each copy is as large as the whole surreal field.

1.2 Arithmetic intersection is not field reconstruction

Theorem B (Transcendental failure of fraction-field descent). *Choose two copies in Theorem A. Then*

$$\text{Frac}(R_\alpha) = K_\alpha, \quad \text{Frac}(R_\alpha \cap R_\beta) \subsetneq K_\alpha \cap K_\beta = L_{\mathbb{R}}.$$

The latter extension is transcendental for every set-sized H , including $H = \{0\}$. For $H \neq 0$, the smaller field already contains every real constant. For $H = \mathbb{Q}$, the element

$$F = 1 + \sum_{n=1}^{\infty} \omega^{-n!}$$

is transcendental over $\text{Frac}(R_\alpha \cap R_\beta)$. The corresponding assertions hold for the Gaussian omnific copies inside $\mathbf{No}(i)$.

This is a failure of a particular descent operation, not a failure of $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. Indeed that classical identity is indispensable to the argument. Each copy separately supplies denominators for every element of its field, but the *shared* denominators need not suffice for their shared field. The phenomenon remains after the missing-real-constants explanation has been removed. Sections 9 and 10 prove the result.

1.3 An ordinary compositum is smaller than a full Hahn join

For full Hahn fields in the same normal-form presentation, their *Hahn join* is the full Hahn field on the sum of their exponent groups. Their *ordinary compositum* is the field generated using finitely many field operations. These should not be conflated.

Theorem C (A transcendental gap above the compositum). *There are explicit strong embeddings Φ_0, Φ_1 as in Theorem A with $H = 0$ for which, writing*

$$K_j = \Phi_j(\mathbf{No}), \quad E = \mathcal{K}_{\mathbb{R}}(G_0 + G_1), \quad M = K_0 K_1,$$

the field E contains an explicit countably supported element S transcendental over M . In particular even the relative real algebraic closure of the ordinary compositum is a proper subfield of the Hahn join. The same S is transcendental over the compositum of the complexified copies inside $\mathcal{K}_{\mathbb{C}}(G_0 + G_1)$.

The two-level normal-form construction is explicit in section 7. The formula for S and the coefficient-field obstruction are proved in section 8. The latter argument is independent of the lacunary-series argument used for Theorem B.

1.4 What is new here, and what is not

The classical inputs are the set-cut property of $\mathbf{No}$, its normal-form presentation, Hahn multiplication and inversion, and the real/algebraic closedness criteria. The group extension argument is a concrete saturation construction, not a new saturation principle. The monomial lifting argument is standard Hahn functoriality. The coset-slice proof is presented as an elementary reusable lemma rather than as a certified new theorem of Hahn field theory.

The relevant repository comparison used the documentation catalogue and the reports on omnific-preserving automorphisms and bounded-support Hahn arithmetic at the pinned snapshot [6, 7, 8]. The former concerns stabilizers and rigidity; the latter already proposes much larger algebraically

independent families over bounded-support fraction fields. We neither repackage those results as discoveries nor use their unrefereed main theorems as black boxes. A self-contained one-witness gap argument is supplied here. In the published literature, the failure of fraction-field equality at set-sized cofinality already appears in L’Innocente–Mantova, Proposition 2.4.5 [3].

The proposed contribution is instead the *simultaneous arrangement*: full copies, a prescribed set-sized full Hahn intersection, strong arithmetic compatibility, independence of every finite subfamily, exact meet diagrams, and the two transcendental descent obstructions inside those arrangements. Targeted literature searches do not certify that this combined formulation is absent from all published or unpublished work. No claim of having audited every report or every later repository revision is made.

2 Foundations, support, and size

2.1 A class framework with set-sized constructions

We work in von Neumann–Bernays–Gödel class theory with Global Choice, with sets satisfying ZFC, as in the foundational convention of [2]. Classes are not members of other classes. Thus an ordinal-indexed family of class functions means one class relation whose sections are the functions. A field with carrier **No** is a *class field*; its elements, finite tuples, polynomial coefficient lists, and the supports of individual elements are sets.

The one surreal-specific order principle needed for the construction is:

$$L, R \subseteq \mathbf{No} \text{ sets, } L < R \implies \text{there is } y \in \mathbf{No} \text{ with } L < y < R. \quad (2.1)$$

Empty sides are permitted. The simplest such y , denoted $\{L \mid R\}$, is uniquely determined. We never apply (2.1) to proper-class sides. This restriction is central, not cosmetic.

Global Choice supplies a set-like class well-order of the surreals: order first by birthday and well-order each set-sized birthday level. It similarly supplies a set-like enumeration of $\mathbf{On} \times \mathbf{No}$. For the latter, group pairs by a bound on both their ordinal coordinate and the birthday of the surreal coordinate, then well-order each new set layer. Each initial segment is a set, and the resulting class order has order type **On**.

The recursion below has a *set as its entire state at every ordinal stage*. For any fixed ordinal, ordinary set transfinite recursion with the fixed class parameters constructs that initial segment. Uniqueness makes these initial segments coherent, and elementary class comprehension collects their stages. This is not an appeal to recursion with a proper-class-valued state at each stage, nor to an unmentioned class truth predicate.

2.2 Full Hahn fields in decreasing-exponent convention

If G is an ordered additive subgroup of **No** and k is a set-sized field, let $\mathcal{K}_k(G)$ denote the formal series

$$\sum_{g \in S} a_g \omega^g, \quad a_g \in k,$$

with reverse-well-ordered set support $S \subseteq G$. For $k = \mathbb{R}$ these are actual surreals. For $k = \mathbb{C}$, taking real and imaginary coefficients identifies them with actual elements of $\mathbf{No}(i)$. The union of the two real supports is again reverse well ordered, so this complexification is exact. For other k the notation is purely formal.

A nonzero series has a largest exponent, and its leading coefficient is nonzero. Addition is coefficientwise and multiplication is convolution. The Hahn support lemma guarantees that the support of a product is allowable and that each product coefficient is a finite sum. Normalizing a nonzero series as $a\omega^g(1 + \varepsilon)$ with all exponents of ε negative gives its inverse by a summable geometric family. Consequently $\mathcal{K}_k(G)$ is a field. These statements are understood locally in set-sized subgroups when G is a proper class.

The standard closedness theorem says that a Hahn field of characteristic zero with divisible exponent group is real closed when its coefficients are real closed, and algebraically closed when its coefficients are algebraically closed [3]. The same conclusion holds for our class fields: the supports of finitely many coefficients of a polynomial generate a set-sized divisible subgroup, and the standard set-sized theorem provides the required roots there. In particular,

$$\mathcal{K}_{\mathbb{R}}(G) \text{ is real closed,} \quad \mathcal{K}_{\mathbb{C}}(G) \text{ is algebraically closed} \quad \text{if } G \text{ is divisible.}$$

If H is a set, then $\mathcal{K}_k(H)$ is a set: it is a subclass of the set of all functions $H \rightarrow k$. In contrast, if G is class-isomorphic to $(\mathbf{No}, +, <)$, then $\mathcal{K}_k(G)$ is a proper class, because it contains the proper class of monomials ω^g .

2.3 Strong summation and finite algebraic independence

Definition 2.1 (Hahn summability). A set-indexed family $(x_j)_{j \in J}$ of Hahn series is summable if the union of its supports is reverse well ordered and every exponent belongs to the support of only finitely many x_j . Its sum is the resulting coefficientwise finite sum at every exponent. A map is *strongly additive* if it sends such families to summable families and commutes with their sums.

The sum of a normal form is strong in this sense. Ordinary analytic convergence of a sequence of real constants is a different notion; it is not used in any proof. In particular strong additivity cannot be replaced by an unqualified assertion of continuity.

Definition 2.2 (Linear disjointness for class fields). Subfields P, Q of one class field are linearly disjoint over $L \subseteq P \cap Q$ if every finite L -linearly independent tuple from P remains linearly independent over Q . A finite family is jointly linearly disjoint if its multiplication map on finite tensors over L is injective. Equivalently, this injectivity holds after restricting each factor to any set-sized subfield containing the finitely many elements under consideration and the base when the base is a set.

All tensor arguments below can instead be read as finite matrix arguments. For a class-sized base, linear independence still quantifies only over a finite tuple of coefficients at a time. “Algebraically independent” also has its usual finite-polynomial meaning. We make no assertion about a proper-class-dimensional tensor object in the category of sets.

3 Independent ordered-group embeddings

The core construction is carried out on the *additive exponent group*, not initially on the field. Its flexibility comes from the ability to realize a set-sized cut while avoiding all previously used values.

Lemma 3.1 (A cut avoiding a set). *Let $L, R, E \subseteq \mathbf{No}$ be sets and suppose $L < R$. There is a surreal $y \notin E$ with $L < y < R$.*

Proof. Let $T = \{e \in E : L < e < R\}$. This is a set and $L \cup T < R$. Choose

$$y = \{L \cup T \mid R\}.$$

Then $L < y < R$. Were $y \in E$, it would belong to T , contradicting $T < y$. This also covers empty L , R , or T . $\square$

Lemma 3.2 (Ordered one-point extension with avoidance). *Let D be a set-sized $\mathbb{Q}$ -vector subspace of $\mathbf{No}$, let $f : D \rightarrow \mathbf{No}$ be an ordered additive embedding, and let E be a set-sized $\mathbb{Q}$ -vector subspace containing $f(D)$. For $x \notin D$, there is an ordered additive embedding*

$$f^+ : D + \mathbb{Q}x \longrightarrow \mathbf{No}$$

extending f and such that $f^+(x) \notin E$.

Proof. Set

$$L = \{f(d) : d \in D, d < x\}, \quad R = \{f(d) : d \in D, x < d\}.$$

These are sets with $L < R$. By lemma 3.1, choose $y \notin E$ realizing this cut. Define

$$f^+(d + qx) = f(d) + qy \quad (d \in D, q \in \mathbb{Q}).$$

The representation $d + qx$ is unique because $x \notin D$, so the formula is well defined and additive. If $q > 0$, positivity of $d + qx$ is equivalent to $x > -d/q$; by the defining cut this is equivalent to $y > f(-d/q)$, hence to $f(d) + qy > 0$. For $q < 0$, the same argument reverses the comparison. The case $q = 0$ follows from the order preservation of f . Thus f^+ preserves strict positivity and is an ordered embedding. Its new value is $y \notin E$. $\square$

Definition 3.3 (Independence over an additive subgroup). Subgroups $(G_i)_{i \in I}$ containing H are *independent over H* if, whenever $J \subseteq I$ is finite and $g_i \in G_i$,

$$\sum_{i \in J} g_i \in H \implies g_i \in H \text{ for every } i \in J.$$

Equivalently the images G_i/H have internal direct sum in the quotient additive group. The quotient here is algebraic; H need not be convex.

Theorem 3.4 (Independent group copies over any set-sized base). *Let $H \subseteq \mathbf{No}$ be a set-sized divisible additive subgroup. There is an ordinal-indexed class family of ordered additive embeddings*

$$h_\alpha : \mathbf{No} \longrightarrow \mathbf{No}$$

such that $h_\alpha|_H = \text{id}_H$ and the ranges $G_\alpha = h_\alpha(\mathbf{No})$ are independent over H .

Proof. A divisible subgroup of the torsion-free additive group of $\mathbf{No}$ is a $\mathbb{Q}$ -vector subspace. Fix the set-like enumeration of $\mathbf{On} \times \mathbf{No}$ described in section 2. We process each requirement (α, x) in that order, meaning that x must eventually be in the domain of the α th partial embedding.

At a stage the state consists of partial ordered $\mathbb{Q}$ -linear embeddings $f_\alpha : D_\alpha \rightarrow \mathbf{No}$ for a set of active indices. Their domains and graphs are sets, they contain the identity on H , and their ranges are independent over H . An inactive index has the implicit partial map id_H ; these implicit copies of H need not be stored separately.

Suppose the next requirement is (α, x) . Activate α if needed. If $x \in D_\alpha$, do nothing. Otherwise let

$$E = \text{span}_{\mathbb{Q}} \left(H \cup \bigcup_{\beta \text{ active}} f_\beta(D_\beta) \right).$$

The state is a set, so E is a set-sized vector space. Apply lemma 3.2 to extend f_α to $D_\alpha + \mathbb{Q}x$, choosing its new value y outside E .

We verify that independence survives. A new finite relation modulo H can involve at most one changed component. Write that component as $f_\alpha(d) + qy$. If $q \neq 0$, the relation would express y as a rational linear combination of old range elements and elements of H , contradicting $y \notin E$. Therefore $q = 0$ and the old independence assertion applies. This proves the invariant at successor stages.

At a limit ordinal, take unions of the compatible graphs. There have been only a set of preceding stages, so the resulting state is still a set. Every finite relation already occurred at an earlier stage, so independence is retained. The set-valued recursion is legitimate in the stated class framework.

Finally define h_α to be the union of its partial graphs over all stages. Each requirement was processed, hence its domain is all of $\mathbf{No}$. Every ordered-additive identity is witnessed at some stage, so h_α is an ordered additive embedding fixing H . Any finite relation among final range elements is also witnessed at one stage. Consequently the final ranges are independent over H . $\square$

Corollary 3.5. *For distinct α, β , one has $G_\alpha \cap G_\beta = H$. More generally, for $\alpha \notin J$,*

$$G_\alpha \cap \left(H + \sum_{\beta \in J} G_\beta \right) = H,$$

where the sum means finite sums, even when J is infinite. Every G_α is a proper subgroup of $\mathbf{No}$.

Proof. The first two assertions are exactly the independence condition. To prove properness, choose $\beta \neq \alpha$ and $x \notin H$. Injectivity and the identity on H imply $h_\beta(x) \notin H$. Thus it cannot lie in G_α , although it lies in $\mathbf{No}$. $\square$

Remark 3.6 (The role of the base). The maps h_α fix H as a set of *exponent values*. The field maps constructed next fix $\mathcal{K}_{\mathbb{R}}(H)$. These are different assertions: for a general subgroup H , it need not be true that $H \subseteq \mathcal{K}_{\mathbb{R}}(H)$. No identification of those two roles is used.

4 Strong field lifts and omnific compatibility

Theorem 4.1 (Monomial lift). *Let $h : \mathbf{No} \rightarrow \mathbf{No}$ be an ordered additive embedding. Then*

$$\Phi_h \left(\sum_g r_g \omega^g \right) = \sum_g r_g \omega^{h(g)} \tag{4.1}$$

defines a real-fixing ordered field embedding of $\mathbf{No}$ with image exactly $\mathcal{K}_{\mathbb{R}}(h(\mathbf{No}))$. It preserves and reflects Hahn summability, and its complexification has image $\mathcal{K}_{\mathbb{C}}(h(\mathbf{No}))$.

Proof. An order embedding sends a reverse-well-ordered set to a reverse-well-ordered set, so the formula defines a surreal. It preserves the leading coefficient and therefore the sign of a nonzero real-coefficient series. Coefficientwise addition is preserved. For multiplication, the identity $h(g + g') =$

$h(g) + h(g')$ identifies each convolution coefficient on the two sides; the support lemma ensures that only finitely many terms contribute to each such coefficient. Since $h(0) = 0$, real constants are fixed. The map is injective by uniqueness of normal form, hence is a field embedding.

For a set-indexed family, the union of image supports is the h -image of the union of original supports. Reverse well-ordering and pointwise finiteness are equivalent under this injection. Thus summability is both preserved and reflected, and the sums agree coefficientwise.

Finally, a series supported in $h(\mathbf{No})$ can be pulled back term by term along the inverse of h on its image. Its support remains a set and is reverse well ordered. This proves that the image is the *full* Hahn field on $h(\mathbf{No})$, not merely the field generated by its monomials. Taking real and imaginary coefficients gives the complex assertion. $\square$

Proposition 4.2 (What the lift remembers). *The map Φ_h preserves the three normal-form projections to positive, zero, and negative exponents. It preserves and reflects membership in $\mathbf{Oz}$, and*

$$\Phi_h(\mathbf{Oz}) = \Phi_h(\mathbf{No}) \cap \mathbf{Oz}. \quad (4.2)$$

It commutes with the omnific floor map and with standard part on finite surreals. Its complexification commutes with conjugation, preserves and reflects $\mathbf{Oz}[i]$, and preserves the real-valued modulus in the sense

$$|\Phi_h^{\mathbb{C}}(z)| = \Phi_h(|z|).$$

Proof. Because h is ordered and additive, $h(g) > 0$, $h(g) = 0$, and $h(g) < 0$ are equivalent to the respective conditions on g . The coefficient at zero is unchanged. An omnific integer is precisely a series supported in $\mathbf{No}^{>0} \cup \{0\}$ with integral constant coefficient, so membership is preserved and reflected. The image formula follows by pulling back a series in the intersection using [theorem 4.1](#).

If $a = \lfloor x \rfloor_{\mathbf{Oz}}$, then $a \in \mathbf{Oz}$ and $a \leq x < a + 1$. Applying Φ_h gives an omnific integer in the corresponding unit interval for $\Phi_h(x)$, hence the unique omnific floor. If $x = r + \varepsilon$ is finite with $r \in \mathbb{R}$ and ε infinitesimal, the image has the same decomposition with infinitesimal part $\Phi_h(\varepsilon)$. This proves the standard-part assertion.

Complex conjugation is coefficientwise. Gaussian omnific membership is omnific membership of both real and imaginary parts. Finally $|z|$ is the unique nonnegative square root of $z\bar{z}$. The image of this equation, order preservation on the real part, and uniqueness of that square root prove the modulus identity. $\square$

Remark 4.3 (A floor boundary that must not be dropped). The positive-exponent truncation plus the ordinary floor of the constant coefficient is not always the omnific floor. For example, if $0 < \varepsilon$ is infinitesimal and $n \in \mathbb{Z}$, then $\lfloor n - \varepsilon \rfloor_{\mathbf{Oz}} = n - 1$. The order-and-uniqueness proof above retains this endpoint case automatically.

Proposition 4.4 (Classification within the monomial-preserving class). *Every strongly additive, real-fixing field embedding $\Phi : \mathbf{No} \rightarrow \mathbf{No}$ that maps each Conway monomial to a Conway monomial is uniquely Φ_h for an ordered additive embedding $h : \mathbf{No} \rightarrow \mathbf{No}$.*

Proof. Define $h(g)$ uniquely by $\Phi(\omega^g) = \omega^{h(g)}$. Multiplication of monomials proves additivity, and injectivity proves injectivity of h . A field embedding between real closed ordered fields preserves positivity: a positive element is a nonzero square. Hence it preserves order. For $g > 0$, the monomial ω^g exceeds every positive real number. Its image has the same property because the reals are fixed. A Conway monomial has this property exactly when its exponent is positive, so $h(g) > 0$. Thus h is

ordered. Applying strong additivity to each normal form gives (4.1). Uniqueness is immediate from the monomials. $\square$

Warning 4.5 (Preserving monomials is not preserving the omega-map). The equality $\Phi_h(\omega^g) = \omega^{h(g)}$ does not say $\Phi_h(\omega^g) = \omega^{\Phi_h(g)}$. In general $h \neq \Phi_h$ as maps on surreal numbers. Nor have we shown compatibility with the Gonshor exponential, the Berarducci–Mantova derivation, birthdays, or simplicity. These extra structures are outside the assertions of this article.

5 Coset slices and linear disjointness

The next argument upgrades an exponent-group intersection calculation to linear disjointness of *full* series fields. The subgroups need not be convex, and no quotient ordering is needed.

Definition 5.1 (Normalized coset slice). Let $H \subseteq G$ be additive subgroups and $g \in G$. For $f = \sum_e a_e \omega^e \in \mathcal{K}_k(G)$ set

$$\lambda_{g,H}(f) = \omega^{-g} \sum_{e \in (g+H) \cap \text{supp}(f)} a_e \omega^e \in \mathcal{K}_k(H). \quad (5.1)$$

The value depends on the chosen representative g by the expected monomial normalization. We do not choose a global transversal of G/H .

Lemma 5.2 (Slice linearity and separation). *Every $\lambda_{g,H}$ is $\mathcal{K}_k(H)$ -linear, and the family of slices separates points of $\mathcal{K}_k(G)$.*

Proof. A subset of a reverse-well-ordered support is reverse well ordered, and translation preserves that property. Thus the definition is valid even when H is not convex. If $a \in \mathcal{K}_k(H)$, a product term with exponent $e + h$ belongs to $g + H$ exactly when e belongs to $g + H$. Therefore slicing the product af gives a times the slice of f . The equality is verified coefficientwise in finite convolution sums. Additivity follows directly. If $f \neq 0$, choose any g in its support. Its slice contains the nonzero coefficient a_g at exponent zero, so the slice is nonzero. $\square$

Lemma 5.3 (A finite separating matrix). *If $f_1, \dots, f_m \in \mathcal{K}_k(G)$ are linearly independent over $L = \mathcal{K}_k(H)$, there are $g_1, \dots, g_m \in G$ such that*

$$A = (\lambda_{g_a,H}(f_b))_{1 \leq a,b \leq m} \quad \text{is invertible over } L.$$

Proof. On the m -dimensional space $V = \text{span}_L(f_1, \dots, f_m)$, the restrictions of the slices have common kernel zero by lemma 5.2. Start with V . If the current common kernel is nonzero, choose a nonzero vector in it and a slice not vanishing on that vector. Intersecting with this new kernel strictly lowers dimension. After at most m steps the common kernel is zero. The resulting linear map from V to at most m copies of L is injective, so exactly m independent coordinate functionals can be used, and their matrix in the stated basis is invertible. This is a finite argument, also for class-sized L . $\square$

Theorem 5.4 (Full Hahn fields are disjoint over their full intersection). *Let G_1, G_2 be additive subgroups of one ordered abelian group and put $H = G_1 \cap G_2$. Inside $\mathcal{K}_k(G_1 + G_2)$,*

$$\mathcal{K}_k(G_1) \cap \mathcal{K}_k(G_2) = \mathcal{K}_k(H),$$

and the two fields are linearly disjoint over $\mathcal{K}_k(H)$. The assertion applies to set or class subgroups, with set-sized individual supports, and to any coefficient field k .

Proof. The intersection identity follows from uniqueness of coefficients and support. Put $L = \mathcal{K}_k(H)$ and let $f_1, \dots, f_m \in \mathcal{K}_k(G_1)$ be L -linearly independent. Choose the invertible slice matrix of lemma 5.3 with representatives $g_a \in G_1$.

On the larger field $\mathcal{K}_k(G_1 + G_2)$ take slices λ_{g_a, G_2} with values in $\mathcal{K}_k(G_2)$. They are $\mathcal{K}_k(G_2)$ -linear. For $f \in \mathcal{K}_k(G_1)$, the identity

$$G_1 \cap (g_a + G_2) = g_a + (G_1 \cap G_2) = g_a + H$$

shows that $\lambda_{g_a, G_2}(f) = \lambda_{g_a, H}(f)$. Suppose $\sum_b b_b f_b = 0$ with $b_b \in \mathcal{K}_k(G_2)$. Applying all of these enlarged slices gives $A(b_1, \dots, b_m)^\top = 0$. Since A is invertible already over L , all b_b vanish. This is precisely linear disjointness. $\square$

Corollary 5.5 (Independence of any finite number of full Hahn fields). *If $(G_i)_{i \in I}$ are independent over H , then the full fields $\mathcal{K}_k(G_i)$ are jointly linearly disjoint over $\mathcal{K}_k(H)$. For $i \notin J$, the field $\mathcal{K}_k(G_i)$ is linearly disjoint over $\mathcal{K}_k(H)$ even from the full field $\mathcal{K}_k(H + \sum_{j \in J} G_j)$.*

Proof. The second assertion follows from theorem 5.4 and the corresponding group intersection. For a finite family, induct on its size. The compositum of the preceding fields is contained in the full Hahn field on the sum of their groups, and the next field is linearly disjoint from that larger field. Extension of scalars over a field preserves an injection of vector spaces. Applying this to the already injective finite-tensor multiplication map proves injectivity for the enlarged family. Equivalently, choose a basis of the finite span of the preceding products and apply linear disjointness to that basis. $\square$

Corollary 5.6 (Polynomial-relation descent). *Let $P = \mathcal{K}_k(G_1)$, $Q = \mathcal{K}_k(G_2)$, and $L = \mathcal{K}_k(G_1 \cap G_2)$. For a finite tuple a from P , let $I_L(a) \subseteq L[X_1, \dots, X_n]$ be its polynomial relation ideal. Then*

$$I_Q(a) = I_L(a) Q[X_1, \dots, X_n].$$

Proof. Take a polynomial relation $F(a) = 0$ with coefficients in Q . A finite L -basis $b_1, \dots, b_r$ for the span of its coefficients gives $F = \sum_j b_j F_j$ with $F_j \in L[X_1, \dots, X_n]$. Linear disjointness is symmetric (as is the finite-tensor criterion), so the b_j remain independent over P . It follows that every $F_j(a) = 0$. Thus F belongs to the extended ideal. The other inclusion is immediate. $\square$

6 Prescribed intersections and Boolean meet diagrams

Proof of Theorem A. Apply theorem 3.4 to obtain (h_α) and their independent ranges G_α , and set $\Phi_\alpha = \Phi_{h_\alpha}$. By theorem 4.1, the image is $K_\alpha = \mathcal{K}_\mathbb{R}(G_\alpha)$. Since h_α is the identity on H , it fixes every monomial with exponent in H , and strong additivity makes Φ_α the identity on all of $L_\mathbb{R}$. Each G_α is proper by corollary 3.5; a monomial with exponent outside it witnesses properness of the corresponding field.

The intersection and independence assertions follow from theorem 5.4 and corollary 5.5. Proposition 4.2 gives the omnific assertions. The same groups and the same slice argument with coefficients in $\mathbb{C}$ give the entire complex package. $\square$

Corollary 6.1 (Fixing every element of a prescribed set). *For every set $A \subseteq \mathbf{No}$ there is a set-sized real closed full Hahn field $L \supseteq \mathbb{R} \cup A$ which is the exact pairwise intersection of an ordinal-indexed family of independent strong omnific-preserving copies of $\mathbf{No}$ fixing L pointwise. For a set $A \subseteq \mathbf{No}(i)$, the analogous statement holds with an algebraically closed $L \supseteq \mathbb{C} \cup A$.*

Proof. Take H to be the $\mathbb{Q}$ -span of the union of the normal-form supports of all elements of A . In the complex case take the real and imaginary supports, equivalently the complex support. Replacement and union make this a set, and rational span remains a set. Then $A \subseteq \mathcal{K}_k(H)$. Apply [Theorem A](#). $\square$

This corollary does *not* assert that the intersection can be prescribed to be the ordinary field generated by A , its real closure, or an arbitrary non-Hahn subfield. It prescribes the full support closure just constructed. The distinction is essential to both the proof and the classification below.

Theorem 6.2 (Exact classification of set-sized intersections in this class). *The set-sized fields that occur as intersections of two images of strong, real-fixing, monomial-preserving self-embeddings of $\mathbf{No}$ are exactly the fields $\mathcal{K}_{\mathbb{R}}(H)$ with $H \subseteq \mathbf{No}$ a set-sized divisible additive subgroup.*

Proof. For necessity, [proposition 4.4](#) writes the two images as full Hahn fields on divisible groups G_1, G_2 . Their intersection is $\mathcal{K}_{\mathbb{R}}(G_1 \cap G_2)$. In a torsion-free divisible ambient group, each divisible subgroup is closed under the uniquely defined rational division; therefore their intersection H is divisible. If the intersection field is a set, its monomials form a set and their exponents, recovered by the inverse of the omega-map on monomials, form a set. Thus H is set-sized. Sufficiency is [Theorem A](#). $\square$

6.1 All intersections in an entire diagram

Fix a set of indices I from the family of [Theorem A](#). For $J \subseteq I$ put

$$G_J = H + \sum_{j \in J} G_j, \quad E_J = \mathcal{K}_k(G_J), \quad G_{\emptyset} = H. \quad (6.1)$$

Every sum in G_J is finite; no infinite additive closure is intended.

Theorem 6.3 (Boolean meet realization). *The assignment $J \mapsto E_J$ is injective and order preserving. For $J, T \subseteq I$,*

$$E_J \cap E_T = E_{J \cap T}.$$

The analogous identity holds for every nonempty set-indexed intersection of subsets of I . Moreover E_J and E_T are linearly disjoint over $E_{J \cap T}$. Their full Hahn join is $E_{J \cup T}$. All E_J are real closed for $k = \mathbb{R}$ and algebraically closed for $k = \mathbb{C}$.

Proof. Independence over H means that an element of G_I/H has a unique finite set of nonzero components in the direct sum of the groups G_i/H . It lies in G_J/H exactly when all of these components have indices in J . This proves all the group intersection identities and $G_J + G_T = G_{J \cup T}$. Uniqueness of normal form transfers the intersection identities to full Hahn fields, and [theorem 5.4](#) proves the asserted linear disjointness.

If $j \in J \setminus T$, choose $g \in G_j \setminus H$; then $\omega^g \in E_J \setminus E_T$. This proves injectivity and reflection of containment. The full Hahn join is, by definition, the full Hahn field on the sum of the two exponent groups, hence is $E_{J \cup T}$. Every G_J is divisible, which gives the closedness assertions. $\square$

Remark 6.4 (The word “join”). The theorem realizes a Boolean meet diagram with *Hahn* joins. It does not identify ordinary field composita with those joins. [Theorem C](#) shows that the difference can already be transcendental for two factors.

Corollary 6.5 (Independent choices and absence of new relations). *For distinct indices $i_1, \dots, i_m$, choose $a_j \in K_{i_j} \setminus L_{\mathbb{R}}$. Then $(a_1, \dots, a_m)$ is algebraically independent over $L_{\mathbb{R}}$. The same assertion holds over $L_{\mathbb{C}}$ for the complexified copies. More generally all polynomial relations across any square of [theorem 6.3](#) descend to its intersection field.*

Proof. A real closed subfield has no proper algebraic extension inside an ordered field, so each a_j is transcendental over $L_{\mathbb{R}}$. In the complex case use algebraic closedness. The polynomial algebras $L[a_j]$ therefore have no relations individually, and their finite tensor product is a polynomial algebra on m variables. Joint linear disjointness makes its multiplication map injective. The last statement is [corollary 5.6](#). $\square$

6.2 A limited model-theoretic conclusion

For each fixed first-order formula in the language of ordered fields, the inclusion of any E_J into **No** is elementary, by quantifier elimination for real closed fields. The corresponding complex statement follows from quantifier elimination for algebraically closed fields of characteristic zero [\[4\]](#). This is a formula-by-formula use of ordinary field theory, not the construction of a class satisfaction predicate.

Nothing here asserts elementary embedding in the expanded language with a predicate for **Oz**, a symbol for the omega-map, or a derivation. The independence proved is algebraic independence and linear disjointness; it is not being silently identified with a model-theoretic independence notion in any richer theory.

7 Explicit copies with intersection exactly the constants

The general construction can fix an arbitrary set-sized Hahn core, but uses class bookkeeping. When $H = 0$ there is a direct formula. It also furnishes the separated scales needed for [Theorem C](#).

Lemma 7.1 (Compression of the surreal order). *The function*

$$\psi(x) = \frac{1}{2} + \frac{x}{2(1+|x|)}$$

*is an order isomorphism from **No** onto the open interval $(0, 1)$.*

Proof. On the nonnegative half-line, $x/(1+x)$ is strictly increasing and belongs to $[0, 1)$. On the nonpositive half-line, $x/(1-x)$ is strictly increasing and belongs to $(-1, 0]$. The formulas agree at zero, proving strict monotonicity and the stated range containment. For $u \in (0, 1)$ put $s = 2u - 1$ and $x = s/(1 - |s|)$. Direct substitution on the two signs of s gives $\psi(x) = u$. All denominators are positive in the ordered field. $\square$

For every ordinal α let

$$I_\alpha = (3\alpha, 3\alpha + 1), \quad \sigma_\alpha(a) = 3\alpha + \psi(a),$$

where the arithmetic in 3α is surreal field arithmetic. These intervals are pairwise disjoint. Indeed $\alpha < \beta$ for ordinals implies $\alpha + 1 \leq \beta$ in the surreal order, hence $3\alpha + 1 < 3\beta$.

Define an *inner* map on normal forms by

$$h_\alpha \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{\sigma_\alpha(a)}. \quad (7.1)$$

This map is an ordered, real-linear, strongly additive *group* embedding. Additivity only uses the injective relabeling of coefficients; no additivity of σ_α is needed. It is not a unital field map:

$$h_\alpha(1) = \omega^{3\alpha+1/2}. \quad (7.2)$$

Its image is precisely the additive class of all surreal normal forms supported in I_α . Pulling back the support along the order isomorphism σ_α proves fullness of that image.

Now apply the *outer* construction of [theorem 4.1](#):

$$\Phi_\alpha \left(\sum_g c_g \omega^g \right) = \sum_g c_g \omega^{h_\alpha(g)}. \quad (7.3)$$

The distinction between (7.1) and (7.3) is crucial: the outer exponents g are themselves surreal numbers with their own normal forms.

Theorem 7.2 (An explicit independent ordinal family). *The maps (7.3) are strong, real-fixing, monomial-preserving proper field embeddings. Their image fields are jointly linearly disjoint over $\mathbb{R}$, have pairwise intersection $\mathbb{R}$, and their omnific image rings have pairwise intersection $\mathbb{Z}$. Their complexifications have the corresponding intersections $\mathbb{C}$ and $\mathbb{Z}[i]$.*

Proof. The inner ranges G_α have disjoint normal-form support intervals. In a finite sum of elements from different ranges, no nonzero coefficient from one interval can be canceled by another interval. Therefore these ranges are independent over zero. Apply [theorem 4.1](#), [proposition 4.2](#), and [corollary 5.5](#). $\square$

For example,

$$\Phi_0(\omega) = \omega^{\omega^{1/2}}, \quad \Phi_1(\omega) = \omega^{\omega^{7/2}}.$$

Both maps fix every real number. Since $\Phi_0(1) = 1$, the first identity also explicitly shows

$$\Phi_0(\omega^1) \neq \omega^{\Phi_0(1)}.$$

Thus preservation of the set of Conway monomials is much weaker than commutation with the omega-map as a unary operation on all of $\mathbf{No}$.

Remark 7.3 (A tempting formula that does not give a proper copy). Replacing every exponent g by cg with a fixed $c > 0$ gives an automorphism, not a proper self-embedding, because $c\mathbf{No} = \mathbf{No}$. The compression above is applied one normal-form level below the field lift; that is what makes the exponent-group image genuinely proper.

8 A transcendental gap above an ordinary compositum

Take G_0, G_1 from [section 7](#). These groups are not merely independent. Their scales are separated:

$$|a| < b/n \quad (a \in G_0, 0 < b \in G_1, n \in \mathbb{N}_{>0}). \quad (8.1)$$

Indeed the inner leading exponent of a nonzero a lies in $(0, 1)$, whereas that of b lies in $(3, 4)$. Leading-monomial comparison proves (8.1). Thus G_0 is convex in the ordered group $G_0 + G_1$, and the sign of $a + b$ for $b \neq 0$ is the sign of b .

8.1 Iterated Hahn coordinates

Lemma 8.1 (Re-expansion at separated scales). *Suppose A, B are additive subgroups with $A \cap B = 0$ and every element of A is infinitesimal relative to every positive element of B , in the sense of (8.1). Then regrouping by B -coordinate gives a field isomorphism*

$$\mathcal{K}_k(A + B) \simeq \mathcal{K}_{\mathcal{K}_k(A)}(B), \quad \sum_{a+b} c_{a,b} \omega^{a+b} \mapsto \sum_b \left(\sum_a c_{a,b} \omega^a \right) \omega^b. \quad (8.2)$$

On the right the notation means iterated Hahn series; the coefficient field may be a class, but each individual series uses only a set of coefficients. The isomorphism is order preserving for ordered k .

Proof. The decomposition $a + b$ is unique, and the order on $A + B$ is lexicographic with B the dominant coordinate. The image in B of a reverse-well-ordered support is reverse well ordered: the inverse image of a nonempty subset of that image has a greatest element, whose projection is greatest in the subset. Each fiber is reverse well ordered in A .

Conversely, a reverse-well-ordered set of B -coordinates, with a reverse-well-ordered support in each fiber, yields a reverse-well-ordered lexicographic union. Its size is a set by replacement and union. The two regroupings are inverse. Addition and multiplication agree coefficientwise: there are finitely many pairs of B -coordinates contributing to each outer product coefficient, and within each resulting coefficient Hahn multiplication in A applies. The dominant B -coordinate and then the leading coefficient in $\mathcal{K}_k(A)$ also give exactly the original order. $\square$

For $k = \mathbb{R}$ or $\mathbb{C}$, put

$$K_0 = \mathcal{K}_k(G_0), \quad K_1 = \mathcal{K}_k(G_1), \quad E = \mathcal{K}_k(G_0 + G_1), \quad M = K_0 K_1.$$

In the iterated coordinates, K_0 is the coefficient field and K_1 has coefficients in the original k .

Lemma 8.2 (Finite coefficient-field control). *Every finite subset of the ordinary compositum M is contained, in the iterated coordinates, in a field $\mathcal{K}_D(G_1)$ where $D \subseteq K_0$ is a field finitely generated over k .*

Proof. Each of the finitely many chosen elements is obtained by finitely many field operations from finitely many elements of K_0 and K_1 . Let $a_1, \dots, a_r \in K_0$ include all the first-kind elements used, and put $D = k(a_1, \dots, a_r)$. The elements of K_1 have coefficients in $k \subseteq D$, while each a_j is a constant in the outer G_1 -series coordinate. The field $\mathcal{K}_D(G_1)$ is closed under the required field operations, proving the assertion. $\square$

This lemma does not claim that every series with coefficients in such a D is in M . It gives a necessary coefficient-field constraint, which is all that is needed.

Theorem 8.3 (A coefficient-transcendence obstruction). *Let $s_1, s_2, \dots \in K_0$ be algebraically independent over k , where $k = \mathbb{R}$ or $\mathbb{C}$, and choose $0 < g \in G_1$. The well-defined series*

$$S = \sum_{n=1}^{\infty} s_n \omega^{-ng} \in E$$

is transcendental over M .

Proof. The outer support $\{-ng : n \geq 1\}$ is strictly decreasing, so the series is valid by [lemma 8.1](#). Suppose a nonzero polynomial over M annihilates S . By [lemma 8.2](#), all its coefficients lie in $\mathcal{K}_D(G_1)$ for one finitely generated field $D \subseteq K_0$ over k .

For $k = \mathbb{R}$, let D' be the relative algebraic closure of D in the real closed field K_0 . It is a set-sized real closed field. To see closedness, a square root of a positive element of D' and a root of an odd-degree polynomial over D' can be taken in K_0 ; each is algebraic over D , so belongs to D' . It is a set because a set of polynomials has only a set of finite root sets. For $k = \mathbb{C}$, the same argument with arbitrary polynomial roots gives a set-sized algebraically closed relative closure $D' \subseteq K_0$.

Since G_1 is divisible, the iterated Hahn field $\mathcal{K}_{D'}(G_1)$ is real closed in the real case, and algebraically closed in the complex case. For a proper-class G_1 this follows, as before, by working with the set-sized divisible subgroup generated by any finite collection of coefficient supports. Thus an element of E algebraic over $\mathcal{K}_D(G_1)$ must belong to $\mathcal{K}_{D'}(G_1)$: use relative algebraic closedness in an ordered field in the real case, or algebraic closedness in the complex case.

Applied to S , uniqueness of its outer expansion now gives $s_n \in D'$ for every n . But

$$\text{trdeg}_k D' = \text{trdeg}_k D < \infty,$$

whereas the s_n are an infinite algebraically independent family over k . This contradiction proves the theorem. $\square$

Explicit witness and proof of [Theorem C](#). Take

$$g_n = h_0(\omega^n) = \omega^{\psi(n)}, \quad g = h_1(1) = \omega^{7/2}, \quad s_n = \omega^{g_n} \quad (n \geq 1).$$

The g_n are $\mathbb{Q}$ -linearly independent: they are distinct inner Conway monomials, so uniqueness of normal form precludes a finite rational linear relation. Consequently the s_n are algebraically independent over $\mathbb{R}$, and also over $\mathbb{C}$ in the complexified field. Indeed the distinct monomials of a polynomial in the s_n acquire distinct exponents $\sum_n m_n g_n$, and no nonzero coefficient can cancel.

The promised actual normal form is

$$S = \sum_{n=1}^{\infty} \omega^{\omega^{\psi(n)} - n\omega^{7/2}}. \tag{8.3}$$

Its exponents strictly decrease: the difference between consecutive exponents is $g + g_n - g_{n+1} > 0$ by scale separation. Thus (8.3) is a legitimate countably supported surreal, not a formal expression awaiting an analytic interpretation. It belongs to E and is transcendental over the appropriate ordinary compositum by [theorem 8.3](#). Transcendence also excludes membership in its relative algebraic closure. $\square$

Remark 8.4 (Why the infinite list of coefficients matters). A rational expression in the two fields can use arbitrary elements of each field, but only finitely many of them. In the separated-scale expansion this bounds the transcendence degree of its coefficient field over the constants. Algebraic extension does not remove that finite bound. The series (8.3) defeats it by displaying infinitely many independent coefficients. This argument is different from simply observing that a series is an infinite sum; many infinite sums, such as a geometric series, already belong to an ordinary compositum.

9 Omnific intersections and their fraction fields

For $k = \mathbb{R}, \mathbb{C}$ write $D_{\mathbb{R}} = \mathbb{Z}$ and $D_{\mathbb{C}} = \mathbb{Z}[i]$, and define

$$\Pi_k(H) = \{x \in \mathcal{K}_k(H) : \text{supp}(x) \subseteq H^{>0}\}, \quad \mathcal{A}_k(H) = D_k \oplus \Pi_k(H).$$

The direct sum is additive; multiplication makes $\Pi_k(H)$ a nonunital ring and an ideal in $\mathcal{A}_k(H)$. In particular,

$$\mathcal{A}_{\mathbb{R}}(H) = \mathcal{K}_{\mathbb{R}}(H) \cap \mathbf{Oz}, \quad \mathcal{A}_{\mathbb{C}}(H) = \mathcal{K}_{\mathbb{C}}(H) \cap \mathbf{Oz}[i].$$

We use the ordinary class fraction field of a class domain to mean the subfield of the ambient field consisting of ratios of two of its elements, with nonzero denominator. No quotient set of a proper class is invoked.

Lemma 9.1 (Common denominators for a set of surreals). *Every set $X \subseteq \mathbf{No}$ has a positive monomial $d \in \mathbf{Oz}$ such that $dX \subseteq \mathbf{Oz}$. The same holds for a set of surcomplex numbers with $\mathbf{Oz}[i]$ in place of $\mathbf{Oz}$, and a positive real monomial denominator. In particular*

$$\text{Frac}(\mathbf{Oz}) = \mathbf{No}, \quad \text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}(i).$$

Proof. The union S of the supports of elements of X is a set. Choose a surreal $a > 0$ with $a > -s$ for every $s \in S$, using the cut property. Then $d = \omega^a$ and every product dx have exclusively positive exponents, except that dx may be zero. Thus they lie in the required integer ring. In the complex case use the union of the real and imaginary supports. Applying the assertion to a singleton gives the fraction-field identities. These identities are classical; compare [3, Proposition 2.4.5]. $\square$

Transporting the lemma through an embedding gives more than merely one denominator for each individual element.

Proposition 9.2 (Separate uniform denominators, no shared uniform one). *Let $K_{\alpha}, R_{\alpha}, L_{\mathbb{R}}$ be as in Theorem A. Then $\text{Frac}(R_{\alpha}) = K_{\alpha}$, and there is a nonzero $d_{\alpha} \in R_{\alpha}$ with*

$$d_{\alpha}L_{\mathbb{R}} \subseteq R_{\alpha}.$$

For two distinct indices, no nonzero element of $R_{\alpha} \cap R_{\beta}$ can satisfy both such containments. The same conclusions hold in the complex and Gaussian setting.

Proof. The image under Φ_{α} of a ratio of omnific integers is a ratio of elements of R_{α} , and every element of K_{α} arises this way by lemma 9.1. Since $L_{\mathbb{R}}$ is a set and is fixed pointwise, apply that lemma to $L_{\mathbb{R}}$ and transport its denominator by Φ_{α} . This yields the uniform denominator d_{α} .

A hypothetical common nonzero denominator d belongs to $R_{\alpha} \cap R_{\beta} \subseteq L_{\mathbb{R}}$. If both containments held, then $dL_{\mathbb{R}} \subseteq \mathcal{A}_{\mathbb{R}}(H)$. But $d \neq 0$ is in the field $L_{\mathbb{R}}$, so $dL_{\mathbb{R}} = L_{\mathbb{R}}$, which is not contained in its omnific part (it contains $1/2$). This is impossible. The same proof uses $1/2 \notin \mathbb{Z}[i]$ in the complex case. $\square$

This proposition already separates two notions of a denominator. To prove Theorem B, however, we need an element with *no shared denominator at all*, not merely the absence of a shared denominator uniform on the entire common field. That stronger assertion is proved next.

9.1 The bounded-support field is exactly the shared fraction field

For this subsection and [section 10](#), use the increasing-support Hahn convention

$$t^h = \omega^{-h}.$$

As $H = -H$, this changes the presentation of $\mathcal{K}_k(H)$, not its underlying field. The support of a nonzero t -series is well ordered and has a least element, its valuation v . Set $v(0) = +\infty$. Let

$$\mathcal{B}_k(H) = \{f \in k((t^H)) : \text{supp}_t(f) \text{ is bounded above in } H\}.$$

It is a subring: finite unions of upper bounds bound sums, and the sum of two upper bounds bounds a product. Its elements can have infinite support inside a bounded interval. In this convention $\mathcal{A}_k(H)$ consists of series with negative exponents and a constant coefficient in D_k .

Proposition 9.3 (Identification of the fraction fields). *If $H \neq 0$, then*

$$\text{Frac } \mathcal{A}_k(H) = \text{Frac } \mathcal{B}_k(H). \quad (9.1)$$

In particular this fraction field contains k and every single monomial t^h with $h \in H$.

Proof. One has $\mathcal{A}_k(H) \subseteq \mathcal{B}_k(H)$. Conversely, take $f \in \mathcal{B}_k(H)$ and choose an upper bound $u \in H$ of its support. Choose $a \in H$ with $a > u$ and $a > 0$, possible in any nonzero ordered abelian group. Both t^{-a} and $t^{-a}f$ have strictly negative support, so belong to $\mathcal{A}_k(H)$. Their ratio is f , giving the reverse inclusion of fraction fields. Constants and monomials have bounded support. $\square$

The identity explains why a mere missing constant is not the obstruction for nonzero H . For example, $r = (r\omega^h)/\omega^h$ for any $r \in \mathbb{R}$ and any positive $h \in H$. The unresolved issue is the unboundedness of supports *within the fixed set group H* .

10 A self-contained cofinal gap argument

The mechanism in this section belongs to bounded-support Hahn arithmetic, not to the new embedding construction. It is included in full so that the transcendental conclusion in [Theorem B](#) does not depend on an unreviewed repository result. The repository's bounded-support report gives stronger independent-family assertions [\[8\]](#), and the non-fraction obstruction is already in [\[3\]](#).

Lemma 10.1 (Cofinal sequences with polynomial gaps). *Let $H \neq 0$ be a set-sized ordered abelian group and put $\kappa = \text{cf}(H)$. There is a strictly increasing cofinal family $(a_\alpha)_{\alpha < \kappa}$ in $H^{>0}$ such that for every positive integer d and every $c \in H$,*

$$a_{\alpha+1} - da_\alpha > c \quad \text{for all sufficiently large } \alpha < \kappa. \quad (10.1)$$

Proof. The cofinality of a nonzero ordered group is an infinite regular cardinal: a finite subset is bounded, and a cofinal union of fewer than κ subsets of size less than κ would contradict minimality of κ by taking bounds for those subsets. Fix a cofinal list $(u_\alpha)_{\alpha < \kappa}$.

When $\kappa = \aleph_0$, choose recursively $a_0 > 0, u_0$ and

$$a_{n+1} > u_{n+1}, \quad a_{n+1} > (n+2)a_n.$$

The list is cofinal. For fixed d , once $n + 2 \geq d + 1$ the difference in (10.1) exceeds a_n , and this eventually exceeds c .

When $\kappa > \aleph_0$, at stage α the set

$$\{0, u_\alpha\} \cup \{na_\beta : \beta < \alpha, n \in \mathbb{N}_{>0}\}$$

has cardinality less than κ and hence is bounded above in H . Choose a_α strictly above it. The resulting family is increasing and cofinal. At a successor stage, $a_{\alpha+1} > (d+1)a_\alpha$ for every positive integer d . Thus again the difference in (10.1) exceeds a_α and is eventually larger than c . Since an infinite cardinal has no last ordinal below it, all the successor indices used are still below κ . $\square$

Theorem 10.2 (One transcendental series over all bounded-support fractions). *For any field k and nonzero set-sized ordered abelian group H , choose a sequence from lemma 10.1. Then*

$$F = 1 + \sum_{\alpha < \kappa} t^{a_\alpha} \quad (10.2)$$

is transcendental over $\text{Frac } \mathcal{B}_k(H)$.

Proof. The support is a well-ordered set, so F is a Hahn series and $v(F) = 0$. Suppose a nonzero polynomial over $\text{Frac } \mathcal{B}_k(H)$ annihilates F . Clearing its finitely many denominators gives

$$Q(Y) = \sum_{j=0}^d b_j Y^j \in \mathcal{B}_k(H)[Y] \setminus \{0\}, \quad Q(F) = 0,$$

where $d \geq 1$. Each nonzero coefficient has support bounded both below and above: below by its least exponent and above by the definition of $\mathcal{B}_k(H)$. Choose common bounds $b, c \in H$ for the supports of all nonzero coefficients, with $b \leq c$.

For $\alpha < \kappa$ let

$$P_\alpha = 1 + \sum_{\beta \leq \alpha} t^{a_\beta}.$$

These are distinct elements of the same Hahn field, including at limit indices; they need not be polynomials in t . Their supports lie in $[0, a_\alpha]$, and

$$v(P_\alpha) = 0, \quad v(F - P_\alpha) = a_{\alpha+1}.$$

Finite Hahn multiplication gives

$$\text{supp}_t Q(P_\alpha) \subseteq [b, c + da_\alpha]. \quad (10.3)$$

The elementary factorization

$$F^j - P_\alpha^j = (F - P_\alpha) \sum_{\ell=0}^{j-1} F^{j-1-\ell} P_\alpha^\ell$$

and the nonnegative valuations of the factors in the finite sum give

$$v(Q(F) - Q(P_\alpha)) \geq b + a_{\alpha+1}. \quad (10.4)$$

By lemma 10.1, eventually

$$b + a_{\alpha+1} > c + da_\alpha.$$

Since $Q(F) = 0$, (10.4) says that every nonzero exponent of $Q(P_\alpha)$ is above the upper bound in (10.3). Therefore $Q(P_\alpha) = 0$ for every sufficiently large α . This gives infinitely many distinct roots of the nonzero finite-degree polynomial Q , a contradiction. The argument uses no characteristic-zero assumption. $\square$

Corollary 10.3 (The factorial witness). *In $\mathbb{R}((t^{\mathbb{Q}}))$ or $\mathbb{C}((t^{\mathbb{Q}}))$, the series*

$$F = 1 + \sum_{n=1}^{\infty} t^{n!}$$

is transcendental over the fraction field of all bounded-above-support series, hence over $\text{Frac } \mathcal{A}_k(\mathbb{Q})$.

Proof. The positive integers $n!$ are cofinal in $\mathbb{Q}$ and satisfy $(n+1)! - dn! = (n+1-d)n! \rightarrow +\infty$ for each fixed d . Thus the proof of [theorem 10.2](#) applies to this explicit sequence. For clarity, its partial sums start with $1+t$, $1+t+t^2$, $1+t+t^2+t^6$; there is no duplicate exponent from a $0!$ term. $\square$

Proof of Theorem B. By [Theorem A](#), the common field is $L_k = \mathcal{K}_k(H)$ and the common integer ring is $\mathcal{A}_k(H)$. Each full integer copy has fraction field equal to its field copy by [proposition 9.2](#). If $H \neq 0$, combine [proposition 9.3](#) and [theorem 10.2](#) to obtain an element of L_k transcendental over $\text{Frac } \mathcal{A}_k(H)$. If $H = 0$, the common ring is $\mathbb{Z}$ or $\mathbb{Z}[i]$ and its fraction field is $\mathbb{Q}$ or $\mathbb{Q}(i)$. The ambient constant field $\mathbb{R}$ or $\mathbb{C}$ contains a transcendental element over that countable field: its algebraic elements are a countable union of finite root sets, whereas the constant field is uncountable. Finally [corollary 10.3](#) supplies the specified witness for $H = \mathbb{Q}$. This proves every assertion. $\square$

Corollary 10.4 (Failure witnessed by the denominator equation). *For $H = \mathbb{Q}$ and F from [corollary 10.3](#), there are nonzero $d_\alpha \in R_\alpha$ and $d_\beta \in R_\beta$ with $d_\alpha F \in R_\alpha$ and $d_\beta F \in R_\beta$, but there are no $a, d \in R_\alpha \cap R_\beta$ with $d \neq 0$ and $dF = a$.*

Proof. The separate denominators follow even uniformly from [proposition 9.2](#). A shared solution would express $F = a/d$ in $\text{Frac } \mathcal{A}_{\mathbb{R}}(\mathbb{Q})$, contradicting transcendence. The same proof works for the Gaussian rings. $\square$

10.1 Where the two transcendence arguments differ

The gap above an ordinary compositum in [section 8](#) is detected by an infinite algebraically independent *coefficient* family in a separated scale decomposition. The gap above the shared fraction field in this section is detected by a cofinal sequence of *support exponents*, even with all nonzero coefficients equal to one. Neither proof is a cardinality-only argument, and neither concludes merely that an element lacks a particular chosen representation: each excludes every finite-degree algebraic relation over the stated base.

11 A set-sized version of the construction

The class construction has a direct conditional set-sized analogue. This both isolates the exact saturation used and gives a more conventional target for formalization.

Definition 11.1. For an uncountable regular cardinal κ , a linear order is an η_κ -order if every cut $L < R$ with $|L \cup R| < \kappa$ has a strict intermediate point. Empty sides are allowed.

Theorem 11.2 (Independent embeddings in a saturated ordered group). *Let κ be uncountable regular, and let V be a divisible ordered abelian group of cardinality κ whose order is an η_κ -order. If $H \subseteq V$ is a divisible subgroup of cardinality less than κ and $2 \leq |I| \leq \kappa$, then there are ordered additive proper embeddings $h_i : V \rightarrow V$ fixing H whose ranges are independent over H .*

Proof. Enumerate the requirements $I \times V$ in κ stages. Before each stage, fewer than κ elements and partial graph entries have been introduced. Their rational span still has size less than κ ; uncountability and regularity ensure this also at limit stages below κ .

The one-point proof works verbatim. To avoid the current span E , form the set T of its elements realizing the current cut and realize the enlarged cut $L \cup T < R$. All these sets have size less than κ , so the η_κ hypothesis provides the new value. Independence is preserved by exactly the same finite-relation argument. At the end all domains are V , and another independent range witnesses properness of each one. $\square$

This theorem is conditional on the existence of the stated group; it does not assume that every cardinal admits such a structure. The uncountability hypothesis is purposeful: in a countable group, adjoining one rational vector already gives a countable domain, so mere realization of finite order cuts would not justify the same proof.

Lifting these embeddings to full set-sized Hahn fields gives the same intersection and disjointness statements. Those Hahn fields need not have cardinality κ ; the theorem places no unwarranted cardinality bound on all their allowable supports.

12 Further research questions

The questions below are proposed by the present theorems. They are not presented as a list of previously published open problems, nor as claims that their answers must be difficult. The first four most directly extend the proved results.

Research question 1 (Intersections outside the full Hahn category). Which set-sized real closed subfields $L \subseteq \mathbf{No}$ containing $\mathbb{R}$ can occur as the exact intersection of two strong, omnific-preserving self-embedding images if the monomial-preserving hypothesis is removed?

Why it is the next step. [Theorem 6.2](#) answers the monomial-preserving case exactly. A non-Hahn example, or an obstruction forcing Hahn closure for a larger class of embeddings, would identify the precise information carried by the chosen monomial cross-section. Strongness alone should not be assumed to settle it.

Research question 2 (Relative classification of pairs of copies). Suppose two independent pairs have the same core $\mathcal{K}_{\mathbb{R}}(H)$. Which invariants determine whether an omnific-preserving automorphism of $\mathbf{No}$ fixing that core carries one pair to the other?

Concrete first case. Compare pairs whose exponent groups form different ordered amalgams over H , even when their abstract additive quotients agree. The ambient ordering of mixed sums is an evident invariant for monomial lifts, but it is not proved here to be complete.

Research question 3 (An exact description of ordinary composita). For independent groups G_1, G_2 over H , characterize the subfield $\mathcal{K}_k(G_1)\mathcal{K}_k(G_2)$ inside $\mathcal{K}_k(G_1 + G_2)$. Determine the transcendence degree of the full Hahn join over the ordinary compositum in natural set-sized and class-sized examples.

Available starting point. In the separated case, [lemma 8.2](#) gives a necessary finite coefficient-field condition, and [theorem 8.3](#) supplies a transcendental obstruction even after algebraic closure. A necessary-and-sufficient condition would be much stronger. For a proper-class extension, formulate dimension statements as independent families of specified set cardinalities rather than casually assigning a set cardinal to the whole class.

Research question 4 (Descent for the arithmetic intersection). How do integral closure, complete integral closure, and finite-type algebraic geometry over $\mathcal{A}_k(H)$ compare with those over $\text{Frac } \mathcal{A}_k(H)$ and $\mathcal{K}_k(H)$? Which descent failures besides the single denominator equation are forced by the transcendental gap?

Important boundary. The elementary fraction-field failure proved here does not by itself calculate a normalization, a Picard group, or a nontrivial projective module. Those require additional arguments and should be compared with the repository’s dedicated arithmetic reports.

Research question 5 (Large fixed bases). For which proper-class divisible subgroups $H \subsetneq \mathbf{No}$ can independent ordered-group copies fixing H be constructed? What cut-size or cofinality condition replaces the set-sized hypothesis of [theorem 3.4](#)?

Obstacle to the present proof. Its first partial domain and its forbidden range span must be sets. A proper-class H invalidates that step. This is a limitation of the argument, not a proof that every such extension is impossible.

Research question 6 (Compatibility with additional surreal operations). Can versions of the prescribed-intersection theorem be proved for embeddings commuting with the Gonshor exponential or with a fixed surreal derivation? What cores would necessarily be closed under those operations?

First necessary checks. If an operation is preserved by both embeddings, their images and intersection must be closed under that operation. The explicit maps of [section 7](#) already fail to commute with the omega-map. Therefore they cannot simply be relabeled as solutions in richer languages. Published work on exponential surreal embeddings provides a relevant comparison [\[5\]](#).

Research question 7 (Simplicity and initiality). Which prescribed intersections remain possible when each image is required to be initial for the simplicity relation, or when a specified bound on birthdays is imposed?

Reason for separating this question. The saturation proof uses the simplest realizer of an enlarged cut to choose one point, but it does not preserve the global simplicity relation. Local use of a simplest number is not an initial-embedding theorem.

Research question 8 (Uniform maximal transcendence at a fixed core). For a fixed nonzero set-sized H , obtain an independently checked sharp calculation of $\text{trdeg}(\mathcal{K}_k(H)/\text{Frac } \mathcal{A}_k(H))$ and relate it to $\text{cf}(H)$ and support cardinality. How much of this calculation can be made uniform across all realized intersection diagrams?

Prior work to retain. This is not an untouched direction: the repository’s bounded-support report already proposes a $2^{\text{cf}(H)}$ -sized independent family and optimal support thresholds [\[8\]](#). A useful next contribution is verification, extension, or a genuinely different uniformity statement, not rediscovery of its one-witness special case.

Research question 9 (Elementary embeddings of the field–integer pair). Can the embeddings be chosen elementary in $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$ while retaining a prescribed set-sized intersection and the stated algebraic independence? What extra restrictions does the integer-ring predicate impose?

What has actually been proved. The field embeddings preserve and reflect membership in $\mathbf{Oz}$ and commute with floor, but these facts do not establish preservation of all formulas with alternating quantifiers. Ordinary real-closed-field elementarity is a weaker assertion.

Research question 10 (A verified set-level kernel). Can the coset-slice lemma, finite separating-matrix argument, and saturated ordered-group extension theorem be formalized as reusable library results, then connected to a universe-aware model of the surreal construction?

Concrete target. The next section gives declarations and obligations at the level of mathematical specifications. A verified finite support computation is not a substitute for these support and class-size proofs.

13 Proof dependencies, verification boundary, and conclusion

13.1 A dependency map

The proof of the main copy theorem follows the chain

$$\begin{aligned} & \text{set-cut property} \implies \text{one-point avoidance} \\ & \implies \text{independent group ranges} \implies \text{strong monomial lifts.} \end{aligned}$$

The upgrade from support intersections to algebraic independence is a separate chain:

$$\begin{aligned} & \text{coset slices} \implies \text{finite separating matrix} \\ & \implies \text{linear disjointness} \implies \text{polynomial-relation descent.} \end{aligned}$$

The arithmetic application adds the common-denominator lemma and the cofinal gap argument. The compositum application instead adds the explicit separated-scale construction, iterated Hahn coordinates, and finite coefficient-field control. Neither application is used to justify the other, and neither is needed to justify the group construction.

13.2 A practical formalization decomposition

A set-level implementation could first express full Hahn series in the usual increasing-support convention and prove the following specifications. The names below are proposed identifiers, not claims that declarations with these names already exist.

- (1) `cosetSlice_linear`: restriction of a Hahn series to one coset of a subgroup, followed by monomial normalization, is linear over the Hahn field on that subgroup. No convexity assumption is permitted.
- (2) `cosetSlice_separates` and `finite_slice_matrix`: the family separates points and yields an invertible finite matrix on a linearly independent tuple.
- (3) `hahnSubfields_linearlyDisjoint`: use the same slice matrix after enlarging the scalar subgroup from the intersection to the second subgroup. This is the main reusable field-theoretic lemma.
- (4) `orderedEmbedding_extend_avoid`: realize a cut of the current rational span while avoiding a prescribed small subspace, with explicit cardinality hypotheses for [theorem 11.2](#).
- (5) `boundedSupport_gap_transcendental`: formalize the two support bounds (10.3) and (10.4), followed by the finite root bound. Start with the factorial sequence over $\mathbb{Q}$ before the transfinite cofinality version.

- (6) `iteratedHahn_coefficientField`: formalize separated ordered sums, the regrouping isomorphism, and finite coefficient-field control of a compositum. This is the kernel of the second transcendence proof.

The final proper-class theorem additionally needs a class or universe framework, a set-like enumeration, and proof that every recursion stage and every support pullback is a set. A set-level Hahn formalization by itself does not discharge that final obligation. The repository contains relevant Hahn and normal-form infrastructure [6], but this article adds no Lean code and makes no machine-verification claim.

13.3 Claims deliberately not made

The following boundaries prevent several plausible but invalid upgrades. The common field is a *full Hahn field* on a *set-sized* subgroup, not an arbitrary prescribed real closed field. Group sums are finite sums, not hidden summable closures. Boolean joins are Hahn joins, not ordinary composita. Strongness concerns Hahn-summable set families, not convergence of all real series. The embeddings preserve omnific membership, not an unproved complete first-order theory of the pair. They are not asserted to preserve the omega-map, exponentiation, derivation, birthdays, or simplicity. The proofs establish the stated theorems but do not establish historical priority or settle a named general factorization conjecture for **Oz**.

13.4 Conclusion

Full surreal copies can be arranged with far more control than a bare universality statement provides. An arbitrary set-sized full Hahn core can be fixed exactly, an ordinal-indexed family of copies can be made independent over it, and their arithmetic parts remain genuine copies of the entire omnific ring. The same construction simultaneously controls the surcomplex field and Gaussian omnific integers.

The two descent failures are complementary. Passing from intersecting omnific rings to their fraction field can lose transcendental elements that belong to both full field copies, even though each copy provides one uniform denominator for the whole common set field. Passing from two full field copies to their ordinary compositum can likewise miss transcendental normal forms allowed by their combined exponent group. Thus support closure, ordinary field generation, and arithmetic localization are three distinct operations. The prescribed-intersection construction makes their difference visible within the actual surreal and surcomplex fields, rather than only in abstract auxiliary examples.

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